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Review article

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Mechanisms of insect respiration

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Abstract

Insect respiration is characterized by the rapid transport of respiratory gases within the organism and efficient exchange with the external environment. The unique respiratory system of insects comprises a network of tracheal tubes that directly supply oxygen to the cells throughout the body of an insect, eliminating the need for blood as an intermediate oxygen carrier. The remarkable diversity of insects and their exceptionally high aerobic scope, possibly the highest in the animal kingdom, demonstrate the success of their respiratory strategy. Microfluidic technology, particularly in the domain of gas microfluidics, also stands to benefit from emulating the mechanical proficiency demonstrated by insects in manipulating fluids at the microscale. Despite this significance, current understanding of the fundamental principles underlying insect respiration is incomplete. This Review presents an overview of insect respiratory physics and identifies promising areas for future investigations.

Sections

Introduction

Gas exchange patterns in insects

Insect respiratory flow physics

Flow control and actuation in insect tracheal networks

Effects of spatial and temporal heterogeneities and asymmetries

Mathematical, computational and microfluidic models of insect respiration

Outlook

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Key points

- Insects achieve some of the highest mass-specific metabolic rates in the animal kingdom owing to their unique, efficient respiratory systems that rapidly transport oxygen directly to the tissues.
- The flow physics of insect respiration is notable because the respiratory airflows of insects involve both low Reynolds numbers and high Knudsen numbers, placing them simultaneously in the slow, creeping and rarefied flow regimes — an unusual intersection.
- The respiratory efficiency of insects may also be attributed to hydrodynamic slip in a substantial proportion of tracheae, which reduces viscous losses.
- Respiratory air flows in the rhythmic tracheal compression regime may be actuated and controlled via a single actuation signal — the abdominal compression frequency of an insect — working in conjunction with the intelligent, distributed, passive collapse mechanics of the tracheal network.
- Mathematical, computational and microfluidic models of insect respiration are essential for deepening our understanding of insect respiratory mechanisms and for translating their highly efficient microscale fluid-handling strategies into technologies.

Introduction

Insects represent the most diverse group of animals on Earth, capable of thriving in varied habitats[1]. After over 480 million years of evolution, they have developed exceptionally efficient respiratory systems, as evidenced by the fact that their metabolic range is the highest in the animal kingdom among an estimated 6–10 million species[2–4]. Their respiratory strategy is fundamentally different from that of other animals, enabling them to meet their energetic demands in various environmental conditions[5–10].

All complex multicellular organisms rely on one of two primary methods for transporting oxygen to the mitochondria of their cells to carry out cellular respiration[11,12]. Most animals, including humans and other vertebrates, rely on lungs for breathing and blood as an intermediary carrier for oxygen in their circulatory systems. By contrast, insects and certain terrestrial arthropods use a system of respiratory pathways called tracheae to transport oxygen-containing air directly to their tissues[13,14]. This latter method offers a clear advantage in speed because the diffusion of respiratory gases in air occurs approximately 10^4 times faster than in blood.

Insects take in air from the external environment through specialized pairs of openings in their exoskeleton called spiracles, which serve as the entrance to their complex tracheal networks. The air is then transported through the ramifying tracheal network to within a few micrometres of virtually every mitochondrion in the body, supplying necessary oxygen for respiration[13,15,16]. Carbon dioxide-rich air is expelled from the body using the same respiratory pathway but in the opposite direction.

Although the basic plan of networked tracheae and tidal air flow is common to all insect respiratory systems (Box 1), the detailed strategies used for ventilation and gas exchange vary vastly among insect species and individuals[17–22]. The diverse tracheal system morphologies and respiratory dynamics found across insect species are closely linked to their life history, habitat and physiological requirements[5,6]. The fundamental design of the insect respiratory system is robust and enables insects to specialize and produce remarkably high metabolic rates relative to their body mass, which in turn support high mechanical power outputs and facilitate flight, a key contributor to their extreme evolutionary diversification[23].

Gaining in-depth insight into the principles of insect respiratory physics has the potential not only to enrich knowledge of insect biology but also to inspire new technologies. For example, emulating the lean, distributed, passive flow control mechanisms observed in insect respiration may lead to the development of lab-on-a-chip microfluidic systems with vastly scaled down control apparatuses, making them truly portable for some applications. Emulating these and other mechanisms insects have evolved to efficiently store, pump, pressurize and eject fluids[24] has the potential to advance various domains such as tissue-on-a-chip or organ-on-a-chip technology, wearable monitoring and medical devices, biotechnology, gas sensing, microscale manufacturing, and microscale chemical synthesis.

Although insect respiration has been studied for over a century, recent advances in imaging techniques have enabled researchers to capture increasing amounts of new data about insect respiratory biomechanics in the past two decades. Much recent work has focused on gathering detailed physiological measurements and on studying the kinematics of the insect respiratory system, both of which have important roles in many aspects of the respiratory process[25]. Non-destructive techniques such as X-ray synchrotron imaging[26–33] and optical coherence tomography[34–37] have provided real-time visualization of the tracheal system in live insects and its fluid dynamics. Scanning electron microscopy and transmission electron microscopy have provided detailed information about the surface morphology and internal structure of tracheae at high magnification[38–41]. Another imaging technique, confocal microscopy, offers high-resolution 3D imaging, enabling researchers to examine the intricate branching patterns and fine structure of tracheae and tracheoles[37,42,43]. These imaging methods, together with an array of mathematical, computational and microfluidic models of insect respiration, have enhanced understanding of the morphology, function and behaviour of the insect tracheal system[14,44–78].

In this Review, we first discuss gas exchange and ventilation patterns in insect respiration. We then turn to insect respiratory flow physics, including questions about the applicability of Murray's law. We outline the flow control and actuation mechanisms used in insect tracheal networks, and the hypothesized effects of spatial and temporal heterogeneities and asymmetries on insect respiratory efficiency. We then present an overview of theoretical and microfluidic models of insect respiration. Finally, we suggest promising avenues for future investigation.

Gas exchange patterns in insects

Insect respiration involves gas exchange, but metabolic demands compete with the need to regulate water loss through the respiratory pathway and manage oxidative stresses in the tissues. Both of these factors depend on the amount of time the respiratory spiracles remain open to the environment. Insects of different species exhibit diverse breathing patterns, which are influenced by a variety of environmental and physiological signals[60,79]. There are three basic spiracular phases in insect respiration: the closed spiracle phase (C-phase), flutter phase (F-phase) and open spiracle phase (O-phase). During the C-phase, spiracles are tightly shut, minimizing gas exchange to limit oxidative stress and

Box 1 | Tracheal system morphology

The morphology of the respiratory system of an insect influences its function, and the fundamental features are shared by most insects. At the same time, variations in tracheal tube size, number, shape and distribution affect respiratory efficiency and metabolic rates[25].

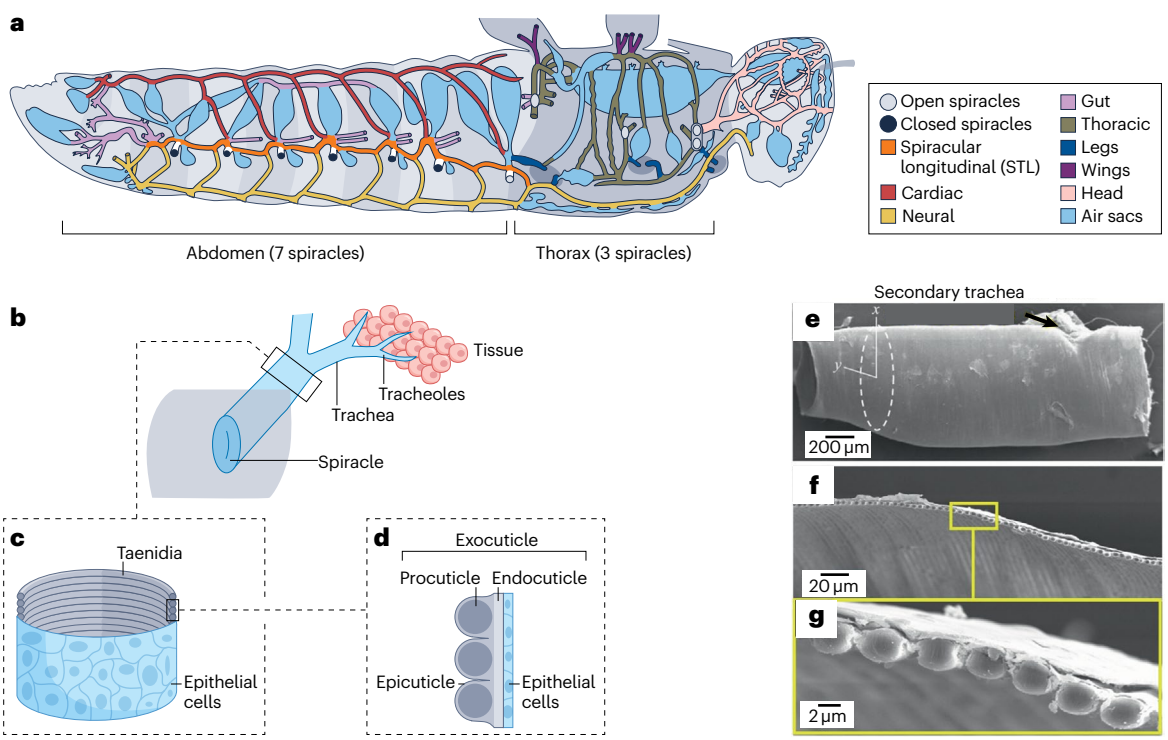
Spiracles, as shown in part **a** of the figure, are the outermost openings of the insect respiratory system. They occur in pairs along the thorax and abdomen[26]. Controlled by two flap-like valves regulated by the contraction and relaxation of small accessory muscles, spiracles open and close to allow air passage (part **b** of the figure)[10,82,144].

Atmospheric air enters the spiracles into longitudinal tracheal trunks (shaded orange, red and yellow in part **a** of the figure) and diffuses through a complex, branching network of thousands to tens of thousands of tracheal tubes. The initial large tracheae subdivide numerous times into smaller tracheal branches and terminate in the smallest and most numerous tracheae, called tracheoles. Here, the atmospheric air meets the body tissues of insects[9,145]. Whereas the diameters of the spiracles and tracheal trunks scale with the size of the insect, tracheole diameters are generally between 0.1–2 μm in diameter in all insects. They have thin walls and moist surfaces that facilitate gas exchange between the air and living cells[62,146,147].

Little general knowledge exists about more detailed morphometric characteristics of insect tracheal networks. Distinct classes of tracheae, based on diameter, have been postulated based on examination of four insect species[148–151]. Five tracheal classes were identified in the stick insect *Carausius morosus*, with diameters ranging from 0.2 to 35 μm, and the tracheal volume was found to comprise about 1.3% of the total body volume of the insect[148]. In several species of spiders that are closely related to insects, three to seven levels of tracheae in these arachnids have been identified[149,151]. Such studies have not been carried out in enough insect species for meaningful conclusions to be drawn about whether insect tracheal networks can be divided meaningfully into generations, as can the branches of the human bronchial tree.

The tracheal tube wall (schematic in part **c** and scanning electron microscopy image from the American cockroach *Periplaneta americana* in part **e** of the figure) has an intricate microstructure, primarily consisting of two layers, the epicuticle and exocuticle (parts **d**, **f** and **g** of the figure)[41,152]. The epicuticle serves as a protective barrier, and the chitin-reinforced exocuticle provides structural support. The exocuticle includes the procuticle and endocuticle, with the procuticle featuring fine geometric thickenings called taenidia.

Taenidia are spirally or circumferentially wound structures along the inner walls that are covered by approximately parallel layers of epithelial cells in the endocuticle. These taenidia reinforce the tracheal walls and are crucial for tracheal reinflation during respiration regimes that involve tracheal collapse[41,94,102]. The outer tracheal surface is lined with epithelial cells that secrete substances during the growth and development of the tracheal system and have a role in the arrangement and organization of taenidia[153,154].



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Some aspects of the taenidial morphometrics of the darkling beetle *Zophobas morio* have been characterized, such as the distribution of taenidial spacing along the tracheal axis[102]. Other studies have shown structural differences in taenidia among different insect stages. For instance, in hawkmoths, butterfly pupae exhibit less prominent, irregularly arranged taenidia compared to those found in the caterpillar stage[31].

Part **a** reprinted with permission from ref. 26, APS. Parts **b**–**g** reprinted with permission from ref. 41, IOP.

water loss, while relying on stored oxygen[21,80,81]. The F-phase involves rapid opening and closing of spiracles in quick succession, facilitating diffusion-based oxygen intake with minimal water loss owing to negative internal pressure[62,82]. In the O-phase, spiracles fully open to release accumulated carbon dioxide, resulting in high water loss and simultaneous oxygen gain[51,83–85].

Based on these spiracular phases, three basic gas exchange patterns have been identified in insect respiration: discontinuous, cyclic and continuous respiration, with distinctive oxygen intake cycles (Fig. 1a). In data from respiration patterns across 19 insect species, supplemented by existing data from 99 other species[86], the three broad patterns can also be distinguished based on the traces of the volume of carbon dioxide production rate $\dot{V}_{CO_2}$ in the different spiracular phases (Fig. 1b–d). According to the study, among 30 insect orders analysed, 5 exhibited discontinuous gas exchange cycle (DGC), 11 showed only cyclic and discontinuous patterns, and 2 displayed only continuous respiration patterns. Data from insects in the remaining 12 orders were inconclusive.

The discontinuous gas exchange cycle

The DGC respiratory pattern involves the lowest rate of gas exchange among the three patterns. It begins with an extended C-phase period, during which the spiracles remain tightly closed, effectively sealing the insect hermetically. This is followed by a short F-phase period during which water loss is kept to a minimum, but carbon dioxide continues to build up. The cycle comes to an end when the amount of carbon dioxide produced by the insect exceeds the buffering capacity of the haemolymph, triggering an O-phase. During the O-phase, the accumulated carbon dioxide rapidly escapes.

Early reports of DGC were in lepidopteran pupae and adult insects of several other species, wherein it was initially described as an extreme form of periodic or intermittent ventilation used while the insects are at rest[25,87,88]. It was identified as a separate, new form of insect respiration after comprehensive investigations on silkworm pupae in the 1960s[89–91]. Early research on adult cockroaches, ants and beetles was based on the idea that DGC evolved to reduce respiratory water loss, consistent with homeostatic requirements[92–94]. Although several competing hypotheses on DGC existed during the mid 2000s because only 51 insect species were recorded to show this pattern[82], in the late 2010s, the number went up to 150 (ref. 95).

Cyclic gas exchange

Cyclic gas exchange involves periodic bursts of gas exchange without prolonged C-phases, with spiracles open for extended periods, resulting in smooth transitions between phases and minimization of water loss.

Rhythmic tracheal compression

Rhythmic tracheal compression (RTC) is a ventilatory pattern that generates convective air flows within the respiratory system via periodic tracheal collapses. RTC is often linked with cyclic gas exchange and thought to be used during periods of high energetic demand. Although the tracheae are reinforced with taenidial thickenings, they are capable of collapsing in some locations because of their thin-walled structure. Synchrotron imaging of many insect species reveals that, during RTC, portions of the tracheal system undergo cyclic compression and reinflation[27,30,96,97]. More specifically, there are four different phases: collapse, in which the tracheal tube rapidly decreases in volume in less than 1 s; static compression, in which the tube is maintained in a compressed state at its minimum volume; reinflation, in which the tube rapidly expands back to its original resting volume, following a time course similar to compression; and inter-compression, the resting period between subsequent compression cycles[98].

Continuous gas exchange

Continuous gas exchange is characterized by the spiracles being either predominantly open or rapidly fluttering; gas exchange rates are high but distinct C-phases and F-phases are absent[86,99]. No discernible rhythmicity is present in this pattern[100], and carbon dioxide release is approximately constant.

Insect respiratory flow physics

An unusual combination of fluid dynamics regimes co-exist in insect respiration. The flow is rarefied and low-speed, with substantial hydrodynamic slip.

Rarefied, low-speed air flow regime

The diameters of insect tracheae range from millimetres in some larger insects, to tens of nanometres at the low end of the range in some smaller insects[39]. Because the length scales involved are so small, the Knudsen number becomes appreciable, and flow within most of the tracheae in insect respiratory systems is in the slip flow regime[71,72,101]. Assuming the commonly used value of 68 nm for the mean free path of air molecules, Knudsen numbers in insect respiratory systems across species range from 0.0001 to nearly 1. These values place insect respiration in the rarefied flow regime, similar to high-altitude airplane flight, flow around satellites and space debris, Knudsen pumps, and microscale or nanoscale devices such as proton-exchange membrane fuel cells.

Additionally, the very small Reynolds numbers achieved, typically much less than unity, places air flow in the insect respiratory system in the viscous-dominated creeping flow regime, similar to the flow of droplets and in capillary tubes, microorganism motion, and lubrication flows. The combination of creeping flow regime and rarefied gas dynamics in insect respiratory systems creates distinct respiratory flow characteristics that are rare in larger organisms.

Hydrodynamic slip

The presence of rarefied flow in insect tracheae means that continuum mechanics assumptions, including that the fluid velocity matches that

of solid boundaries it is in contact with, may not apply. Instead, based on considerations of their size and Knudsen number, the majority of insect tracheae should exhibit hydrodynamic slip, in which gas flow near the walls experiences reduced viscous losses compared to what is seen in flows without slip. This effect is more pronounced in smaller tracheae as the mean free path of gas molecules becomes comparable to the tracheal diameter. For tracheae with diameters below approximately 70 µm, the Knudsen number exceeds 0.001, placing the flow within the slip flow regime according to standard fluid dynamics classifications. Data from at least one beetle species indicate an overlap region in the tracheal network, with diameters ranging from approximately 20–70 µm, wherein they are large enough to collapse rhythmically in response to periodically elevated transmural pressure, yet small enough that they are in the slip flow regime[102].

This reduction in viscous losses owing to hydrodynamic slip (Fig. 2a) is analogous to the Fåhræus–Lindqvist effect observed in the mammalian vasculature (Fig. 2b), wherein the apparent viscosity of blood decreases in the microcirculation. In this region, consisting of blood vessels with diameters of approximately 10–300 µm, the interaction of biconcave red blood cells with strong shear gradients causes the red blood cells to tumble toward the centre of the vessel, forming a cell-free layer (CFL) near the vessel wall. This CFL reduces the effective viscosity of the blood flow, as the red blood cells are confined to a central core region, whereas the CFL region near the vessel wall consists only of plasma, which has a markedly lower viscosity than whole blood. The apparent viscosity of blood decreases in this region with decreasing vessel diameter, reaching a minimum in vessels with diameters of approximately 10–15 µm, which is slightly larger than the diameter of a red blood cell. This region of the vasculature encompasses the majority of blood vessels in the entire vasculature.

Similarly, although through a completely different mechanism, hydrodynamic slip in insect tracheae reduces friction between the gas molecules and the walls, thereby decreasing overall flow resistance. This mechanism enhances transport efficiency in what may be a substantial proportion of the tracheae within the insect respiratory system.

Murray's law and insect respiration

One framework for considering the effects of slip in insect respiratory systems is that of optimally efficient transport through the tracheal network. In mammalian circulatory systems, such optimality results in Murray's law, which describes the approximate blood vessel

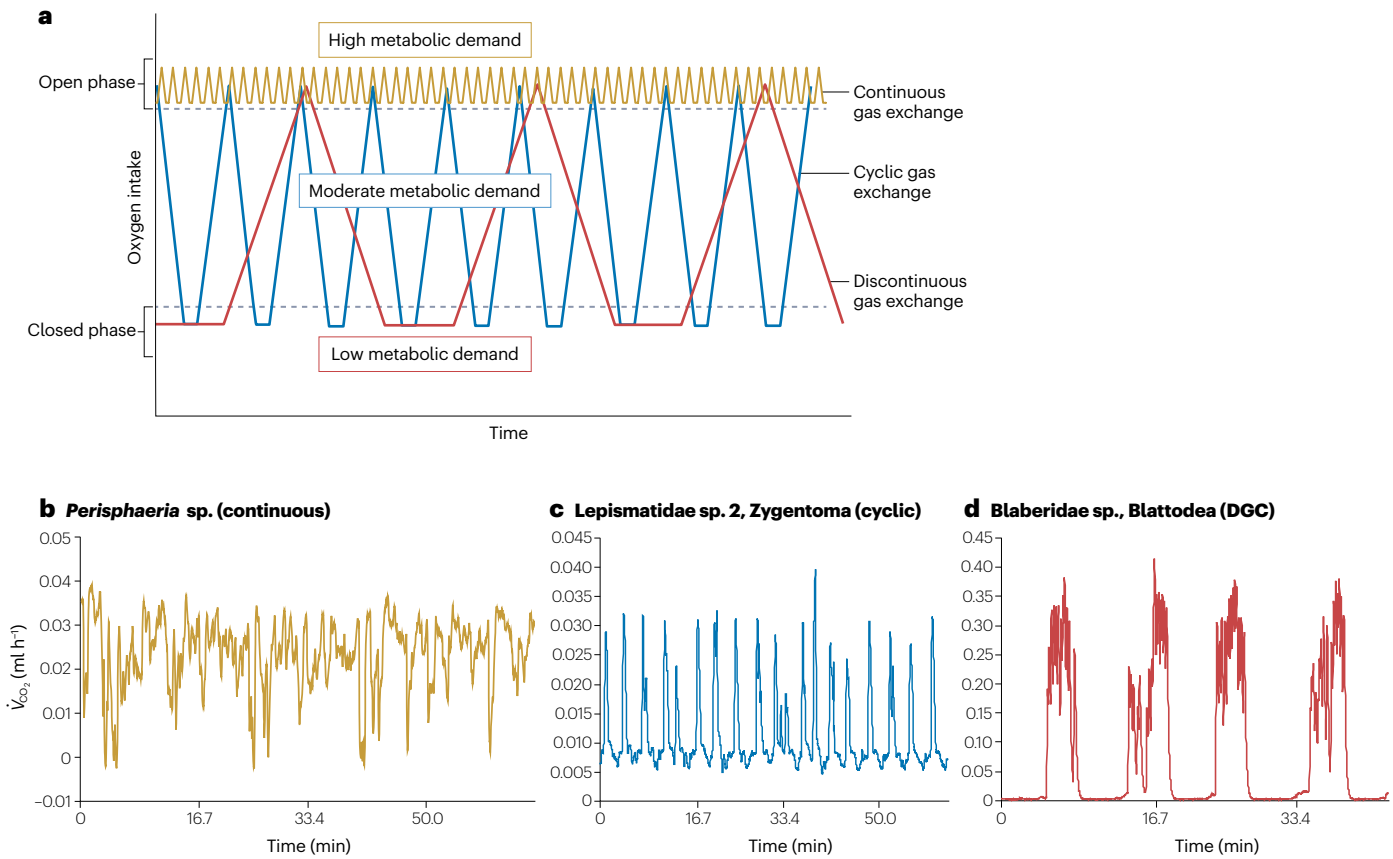

Fig. 1 | The three broad patterns of gas exchange in insects are continuous, cyclic and discontinuous gas exchange. a, In the continuous gas exchange pattern, the spiracles are kept open and oxygen intake is continuous. This pattern is used in times of high metabolic demand. In cyclic gas exchange, the spiracles open and close in a periodic or near-periodic fashion. The cyclic gas exchange pattern is used in times of moderate metabolic demand. In discontinuous gas exchange, the spiracles are closed or fluttering most of the time, with discrete bursts of carbon dioxide release and oxygen intake as needed. This pattern is used in times of low metabolic demand. **b**, A carbon dioxide trace in a cockroach exhibiting continuous gas exchange. **c**, A carbon dioxide trace in a silverfish exhibiting cyclic gas exchange. **d**, A carbon dioxide trace in a giant cockroach exhibiting discontinuous gas exchange. DGC, discontinuous gas exchange cycle. Parts **b**–**d** reprinted with permission from ref. 86, Company of Biologists Ltd.

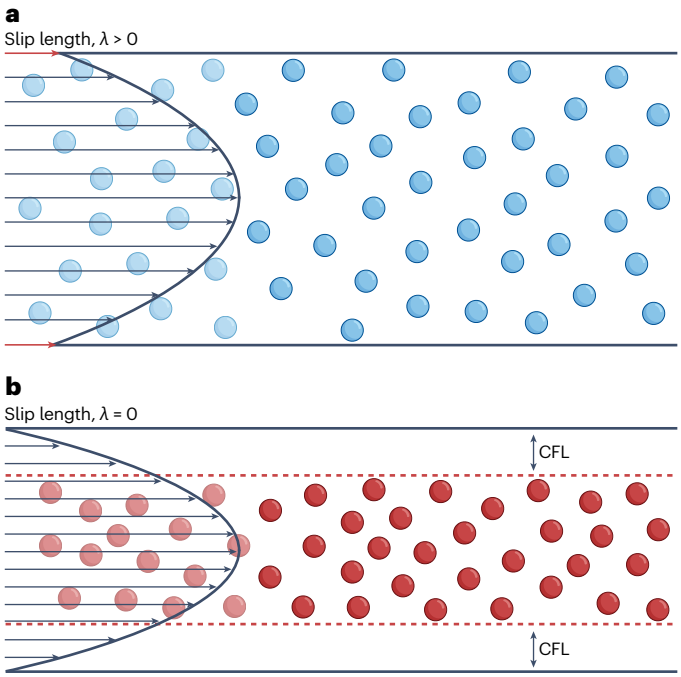

Fig. 2 | Mechanisms exist for reducing viscous losses in both mammalian and insect oxygen delivery pathways. a, In insects, although respiratory gas molecules (blue circles) are present across the entire tracheal cross section, a substantial level of hydrodynamic slip is present in tracheae with diameters below approximately 70 μm. This has the effect of reducing viscous resistance at the tracheal wall in the insect respiratory system. **b**, In mammalian blood vessels with diameters of 10–300 μm, a cell-free layer (CFL) forms near the vessel wall because the oblong (biconcave) red blood cells (red circles) tumble toward the centre of the vessel in response to the radial velocity gradient. The effect of the CFL is to reduce the apparent viscosity of blood in the microcirculation.

branching patterns in the larger arteries, and states that the volumetric flow rate of fluid through a vessel scale as the cube of its radius, $Q \propto R^3$ (refs. 103–108). However, the limited data available suggest that insect respiratory systems do not follow this law[45,109,110]. Murray's law represents the flow rate scaling that minimizes energy losses per unit time in advective transport through ramifying networks and assumes steady, laminar Poiseuille flow with no-slip boundary conditions. However, in insect respiration the mode of transport of respiratory gas molecules transitions from diffusion dominated to advection dominated with increasing tracheal diameter, and advective transport in insect tracheae is modulated by hydrodynamic slip. Furthermore, in larger tracheae in the rhythmic tracheal compression regime, the flow can be unsteady.

By incorporating a standard multiplicative slip factor, $(1 + \frac{4\lambda}{R})$, into the advective transport term, where λ is the slip length, and including a new term that accounts for energy losses owing to diffusive transport, $\pi R^2 \gamma D \Delta C / L$, where γ is a constant associated with the energetic cost of diffusion per mole of the chemical species, D is the diffusion constant of the species, and ΔC is the driving chemical species concentration difference across the tracheal segment of length L, modifications may be made to Murray's law that result in variable scaling predictions for transport in insect tracheae. The classical $Q \propto R^3$ scaling is recovered for large trachea; for smaller tracheae as $R \to 0$, the scaling is $Q \propto R^2$ as predicted in ref. 111. Further assumptions about how the metabolic cost term depends on tracheal geometry, and how features of the tracheal geometry, such as segment length and wall thickness, scale with the tracheal radius, have the potential to yield a wide range of scaling predictions, $Q \propto R^x$, with x ranging from 1 (assuming the tracheal wall thickness is constant and the segment length L scales as $R^{1.5}$ instead of Murray's 'biological scaling' assumption of $L \propto R$) to 4 or 5 (assuming the wall thickness is constant and L scales as R^{-2}), or higher, depending on the assumptions made. Morphometric studies of insect tracheal networks will assist in the development of accurate biophysical scaling laws for insect respiratory networks.

Flow unsteadiness

If one considers the range of Womersley numbers relevant to airflow in large tracheae in the rhythmic tracheal compression regime, assuming tracheal diameters below 200 μm and abdominal compression frequencies below 30 Hz, one finds Womersley numbers between 0 and approximately 0.7. These values imply that airflow pulsatility or unsteadiness induced by tracheal compressions quickly dissipates and the flow even in large insect tracheae can be assumed to be steady away from the compression sites.

Flow control and actuation in insect tracheal networks

In insects – unlike in mammals – tracheal collapse is non-pathological. It appears to be a key element of the efficient, passive, distributed control mechanisms observed in insect respiration. Insect respiratory flow control strategies, described below, appear to allow insects to control flow through thousands of networked tracheal pathways via a single actuation input: the frequency of their abdominal contractions. And in the case of flight, the frequency of wing flapping may take over as the driving actuation frequency.

Tracheal collapse for flow actuation

Unlike in mammals, large insect tracheae routinely experience transmural pressure differences that exceed the critical value at which they collapse. Tracheal collapses, which occur periodically in cyclic respiration regimes, help drive advective airflows through the tracheal system. Insect tracheae are capable of quickly reinflating after collapse and withstanding numerous collapse cycles without mechanical failure. The tracheae are reinforced with helically wound taenidial thickenings, which prevent complete tracheal occlusion and maintain some airflow even in the fully collapsed state. This structural adaptation allows insects to effectively manage internal pressures and ensure adequate oxygen supply to their tissues. The RTC cyclic respiration regime observed in many insect species is a key example of this phenomenon, in which the tracheae cyclically collapse and reinflate, facilitating advective gas transport.

High-resolution imaging techniques, such as X-ray synchrotron imaging, have revealed the kinematics of tracheal collapses in many insect species[26–30]. Figure 3a shows typical X-ray synchrotron images taken during a tracheal collapse study in a beetle. The data from such studies can be digitized for closer analysis (Fig. 3b) and used to develop models of insect respiration (Fig. 3c). Tracheal collapses in the cyclic respiration regime are correlated with the simultaneous, periodic abdominal compressions that are observed in this regime. These compressions are thought to increase the pressure of the haemolymph surrounding the tracheae, which increases the tracheal transmural pressure, leading to localized tracheal buckling across multiple collapse sites, which drives bulk advective flows in many insect respiratory

systems. Microfluidic models of single insect tracheae have shown that an insect-mimetic flow actuation strategy, in which the buckling collapse of discrete segments of the tracheal model wall in response to a periodic transmural pressure signal, can drive a flow through the model trachea if the spatial or temporal symmetry of the collapses are broken in some way, and that the flow rate and direction can be controlled via the actuation frequency alone[73].

Passive, distributed control strategies

Mammals use active, muscularized control mechanisms such as sphincter valves, vasodilation and vasocontraction to direct the flow of oxygen and other nutrients regionally in the circulatory network[112–114]. Insects, by contrast, do not have a means to dilate or constrict their tracheae and lack sphincter-like valves in their (tracheal) microcirculation. Instead, they may use a single actuation signal – their abdominal compressions, the frequency and magnitude of which they can vary actively – in combination with the naturally adaptive, structure-based mechanical properties of their tracheal walls, to direct flow regionally in their respiratory networks.

Microfluidic models of insect tracheal networks provide evidence for this understanding. These models (Fig. 3d) have demonstrated that it is possible to switch the flow through an individual channel in the network off or on via the global actuation frequency alone by applying an insect-mimetic flow actuation strategy based on distributed, localized tracheal buckling in response to a single, global periodic transmural pressure actuation signal[73].

Additionally, the distribution of tracheoles in the respiratory network helps direct flow to high metabolic demand tissues. Tracheoles, which are of diameter 0.1–1 μm and usually have walls that are less than 0.1 μm thick, are the major site of gas exchange between the tissues and the tracheal system[115]. The extensive branching of tracheoles (Fig. 4a) maximizes the surface area for gas exchange, particularly in metabolically active tissues such as flight muscles[115]. Studies on the respiratory system of the migratory locust *Locusta migratoria* have demonstrated that aerobic capacity scales with the surface area of the tracheoles[116,117]. In particular, the flight muscle of these locusts exhibits a 6.1-fold higher aerobic capacity, which correlates with a 6.4-fold greater tracheole inner cuticle surface area

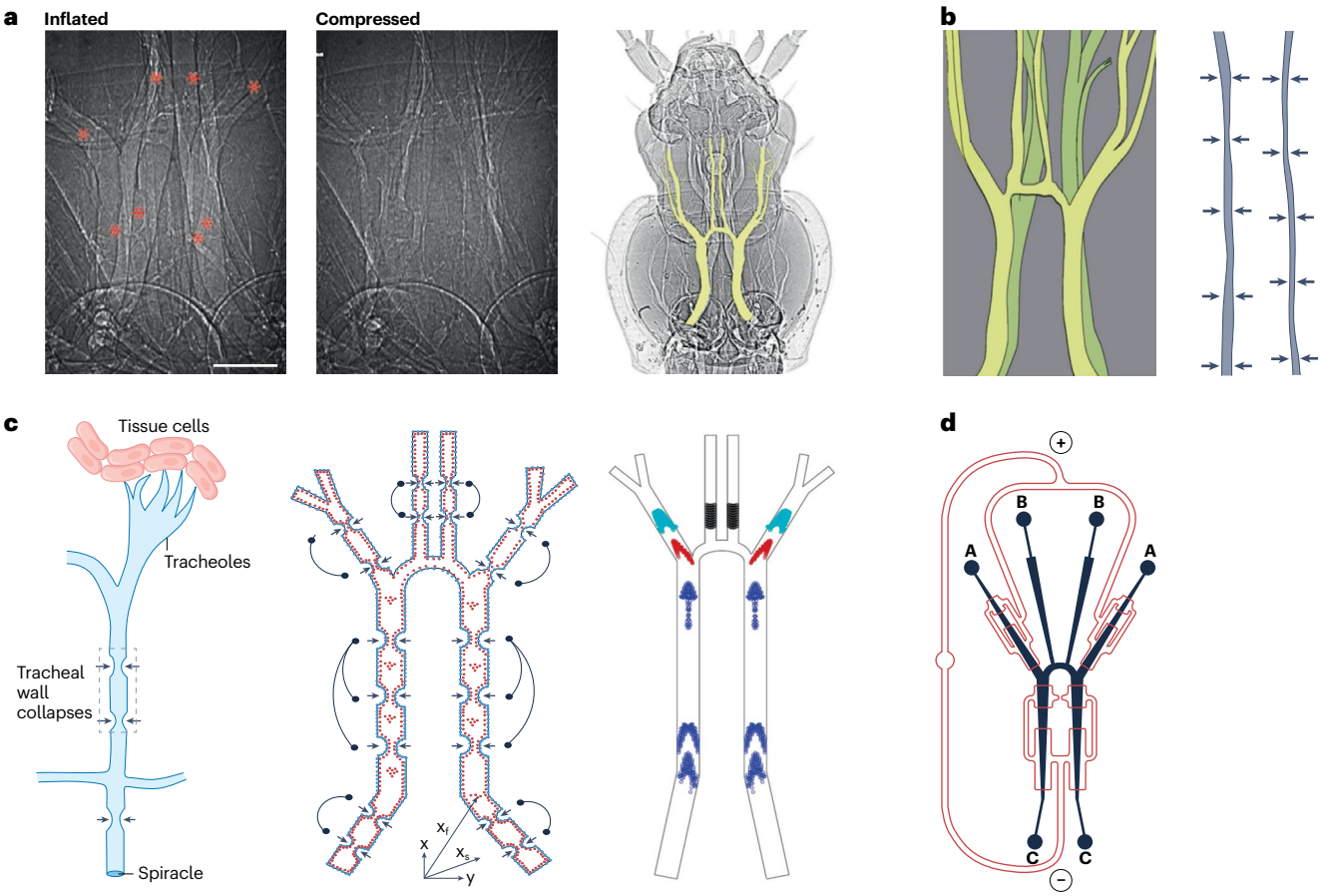

Fig. 3 | Tracheal collapse drives convective airflow in insect respiration. **a**, X-ray synchrotron images of the main thoracic tracheae in a *Platynus decentis* specimen show these tracheae in their fully inflated (left) and fully collapsed (right) states in the cyclic gas exchange respiration regime. Tracheal collapse kinematic data can be digitized (part **b**) and lead to the development of mathematical and computational models of insect-mimetic and insect-inspired tracheal pumps (part **c**). **d**, Several single-channel trachea-mimetic pumps combined into an insect respiratory system-inspired microfluidic network device. The flow in channel B of the device switches on above a critical device actuation frequency of 0.5 Hz. Parts **a**–**c** reprinted with permission from: **a**, ref. 98, Company of Biologists Ltd. (left and middle), ref. 135, Yale Journal of Biology and Medicine (right); **b**, ref. 98, Company of Biologists Ltd. **c**, ref. 64, Virginia Tech. Part **d** adapted with permission from ref. 73, IOP.

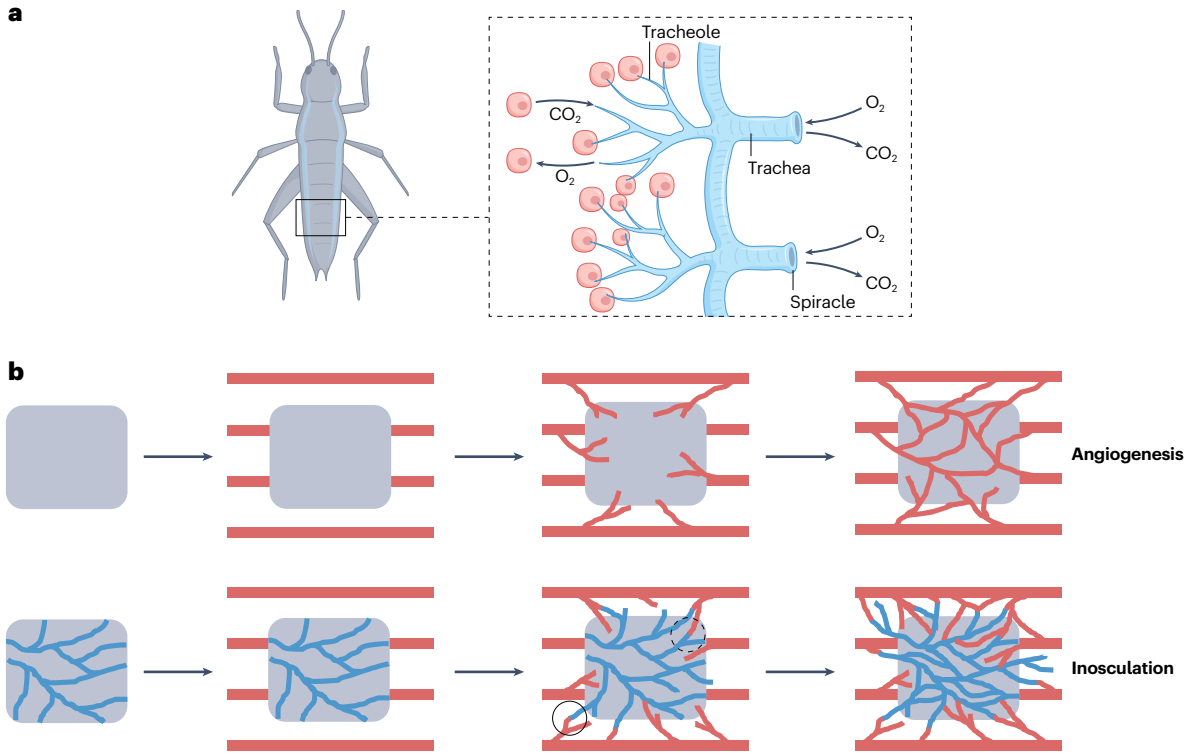

Fig. 4 | The unique respiratory systems of insects transport oxygen directly to tissues via tracheal networks, bypassing the need for blood as an intermediate carrier and thereby achieving some of the highest aerobic scopes and mass-specific metabolic rates in the animal kingdom. a, Insect tracheal networks facilitate oxygen and carbon dioxide exchange using unique tracheal network architectures and flow control strategies. **b**, Engineered tissue vascularization strategies such as angiogenesis and inosculation attempt to optimize the supply of oxygen and other nutrients to the tissues. However, current engineered tissues are limited to thin layers because effective methods for nutrient distribution have not yet been developed. Adopting insect respiratory system-inspired vascularization strategies may help address this roadblock in tissue engineering technology. Part **b** is adapted with permission from ref. 143, Karger Publishers.

compared to the hopping muscle[118]. This correlation underscores the significance of tracheole surface area in supporting high metabolic demands during flight.

In contrast to the passive, elegant and effective flow control mechanisms used in insect respiratory systems, state-of-the-art methods for flow delivery and regulation in engineered microsystems, including very large-scale integrated (VLSI) microfluidic systems, involve the targeted active actuation of pump and valve arrays. Modularizing and miniaturizing these systems is complex and costly owing to the number of accessories needed to power and control the flows. As a result, active flow control strategies are often unsuitable for applications such as implantable microdevices, which need to be small, self-contained and energy-efficient. Implementing insect tracheal network-inspired passive control strategies may allow the next generation of VLSI microfluidic systems to have fine control over the flow through individual branches of the network using a single, global actuation signal alone, thus greatly reducing the required actuation machinery and power requirements and enabling the development of sophisticated, miniature self-contained and implantable VLSI microfluidic devices. Another area in particular need of innovation is that of tissue engineering (Fig. 4b). Ensuring adequate gas and nutrient exchange in engineered or regenerated tissues thicker than 1 mm is difficult using existing techniques[119]. Insect-inspired microscale flow delivery strategies may enhance vascularization and nutrient distribution in engineered tissues, and address this major roadblock in tissue engineering.

Effects of spatial and temporal heterogeneities and asymmetries

Mathematical, computational and microfluidic models of insect tracheae have provided evidence that temporal or spatial asymmetries in the model trachea geometry are necessary to produce a net flow in the model trachea[63,64]. They have further demonstrated that increasing the degree of asymmetry can increase the flow rate produced in the model trachea, up to a point. In such studies, an optimal temporal asymmetry, such as the phase lag between the collapses of a pair of contraction sites, has been observed. For example, when spatial asymmetry was introduced in the length of the collapse sites in a theoretical trachea model, the asymmetric configuration maximized flow produced in the trachea compared to when the collapse sites had the same length[64] (Fig. 5a).

Because respiratory gas molecules act as passive tracers that follow the tracheal air flow, an increase in the advective flow rate produced in a trachea translates to an increase in the rate of molecular clearance or transport. This increase is analogous to observations in mouse respiratory systems, wherein geometric and orientational heterogeneity in airway ciliary arrays was found to enhance particle clearance by breaking up closed orbits and reducing clearance time[120].

Whereas the mechanisms identified for enhancing transport rates in insect respiratory systems rely on asymmetry in the tracheal geometry or timing of localized collapses (Fig. 5b), mice appear to leverage structural heterogeneity (Fig. 5c). Both strategies are based on exploiting the effects of irregularities, and effectively increase fluid-mediated transport in respiratory pathways.

Mathematical, computational and microfluidic models of insect respiration

Predictive models of insect respiration are key to consolidating our understanding of the physical mechanisms at play and translating the knowledge gained about the unique respiratory mechanisms of insects into technological advances. At present, however, models of insect respiration generally either represent very general features of insect respiration or reproduce experimental data taken in highly specific respiration scenarios, usually in a single insect species. Thus, opportunities to improve on current insect respiratory modelling efforts are plentiful. Modelling efforts to date have fallen into three major categories: mathematical models that typically consider only the rate of diffusive transport of oxygen and carbon dioxide into or out of an insect; mathematical and computational models of internal advective airflows in small segments of the insect tracheal system; and microfluidic models that seek to recapitulate the mechanisms insects use to actuate and control flows. We give an overview of each type of modelling effort in the following paragraphs.

Although in reality respiration is accomplished through a combination of diffusive and convective processes within the tracheal system[30,121,122], calculations carried out in the early twentieth century[44–46] suggested that the process of diffusion in itself should be capable of providing an adequate amount of oxygen to meet the needs of the respiring tissues in insects. These calculations, which constitute the first quantitative models of insect respiration, were based on the general 1D diffusion equation expressed as

$$\frac{dC_G}{dt} = -\frac{\kappa_G A p_G}{L},$$

where C is concentration, $\kappa = D \times \alpha$ is the diffusion constant with D the diffusion coefficient and α the solubility, p is partial pressure, L is the length of the diffusive path, A is the area along which diffusion takes place, and G is the gas under consideration. Along with the calculations, different branches of the tracheal system were classified as 'ventilation tracheae' for ventilatory purposes and 'diffusion tracheae' that could be supplied by diffusion alone. Traditionally, it was thought that ventilation tracheae could collapse and pump air, whereas diffusion

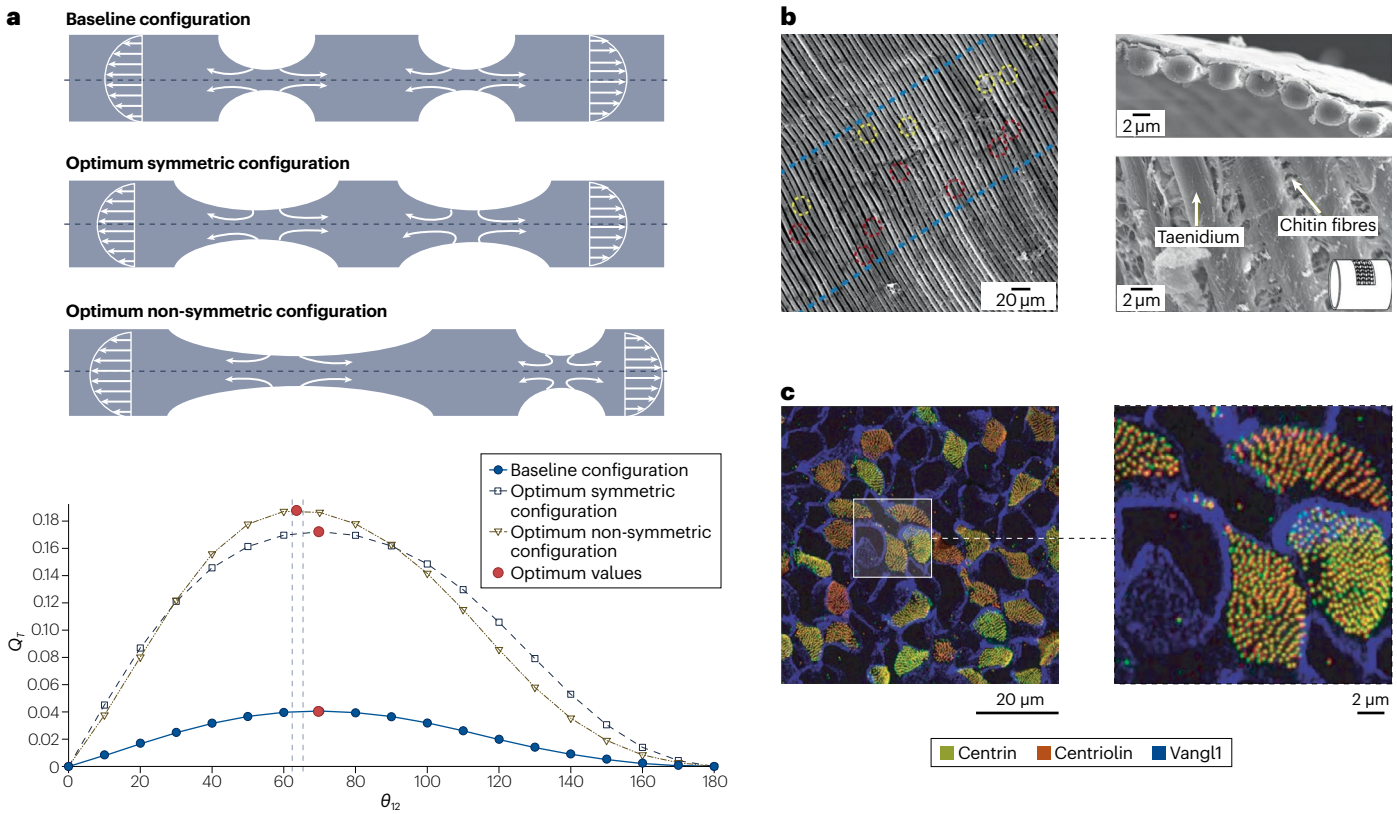

Fig. 5 | Multiscale irregularities and disorder may enhance fluid-mediated transport in insect respiratory systems. a, The baseline, extended symmetric, and asymmetric collapse configurations of a two-collapse mathematical model of an insect trachea (top) and the time-averaged net flow it produces, Q_T, as a function of the periodic collapse timing phase lag parameter, θ_{12} (bottom), for each configuration. $\theta_{12} = 0$ results in no net flow in this model; optimal θ_{12} values are indicated in red. **b**, In insect tracheae, curved airways and non-uniform taenidia (epithelial ridge microstructures) arrangements may enhance particle clearance by breaking axisymmetry and allowing 3D respiratory flow topologies to form **c**, Likewise, in mouse tracheae, multiscale spatial heterogeneities in epithelial ciliary architectures enhance mucus clearance. Part **a** reprinted with permission from ref. 64, Virginia Tech. Part **b** reprinted from ref. 120, Springer Nature Limited. Part **c** reprinted with permission from ref. 41, IOP.

tracheae remained static. Although this work received criticism based on simplified calculations[25] and detailed measurements of diffusion pathways[123,124], the basic conclusions of the model were supported by studies of insect wing movements during flight in the 1960s[125]. Following this early modelling, another mathematical model of insect respiration with a similar diffusion-only approximation for the tracheolar system of insects was developed in the 1970s[14]. Unlike earlier work, this model explicitly represented the limiting case wherein the tracheoles are very small and comparable in diameter to the mean free path of air.

In recent decades, the diffusion-only approximation has been extended to develop several improved models of insect respiration that seek to predict the rate of transport of gas molecules as a function of concentration gradient, geometric parameters and gas properties. In one model, the concentration of respiratory gases is computed in a compartmental gas exchange model[47]; this study was followed by an improved model that distinguished individual spiracular dynamics[48]. Another model computes the diffusive flux of gases by deriving the passage probability for a single particle that diffuses through a stochastically gated boundary switching between open and closed states representing two major spiracular phases[49]. Meanwhile, a two-sensor control system model predicts the concentrations of oxygen and carbon dioxide by two disparate, coupled models that are triggered by each other's threshold concentrations to represent a realistic DGC mechanism[50].

In addition to the models discussed above that consider only diffusive transport, another model incorporated advective effects in the flutter phase of gas exchange[51]. This model assumes a diffusion-based approach to transport for the open and closed respiration phases and pressure-driven Hagen–Poiseuille flow as a convective transport model during the flutter phase[126].

Additionally, diffusion-based models of respiration in aquatic and semi-aquatic insects, which face unique challenges, have been developed[127–130]. Plastron respiration is one of the many unique adaptations in aquatic insects that has been modelled[52,54,131–134]. The gas exchange between the plastron and the surrounding water column can be computed using a model that follows earlier two-component (oxygen and nitrogen)[52] and three-component (oxygen, carbon dioxide and nitrogen)[53] gas exchange models for similar insects.

Diffusion-only models of insect respiration are particularly inaccurate for representing advective transport by intra-tracheal airflow, which has a major role in respiration in many insects[135]. Moreover, gas transport within insect respiratory systems can be unsteady because of abdominal contractions and the resultant rhythmic collapse of tracheae in the RTC respiration regime[48,136]. To better represent unsteady and advective transport, several theoretical and numerical models have been developed to represent air flows in insect tracheae with moving boundaries that imitate such physiological systems[60–62]. A series of mathematical and computational studies were conducted on valveless pumping in the RTC respiration regime in single insect trachea and in tracheal networks, in which realistic tracheal wall motions were included[63–70]. Although these models address some of the complexities of the geometry and dynamics of the tracheal system, the influence of hydrodynamic slip was not taken into account. In later models, the wall contraction-driven 2D framework of ref. 63 was adapted to investigate the effects of slip[71]. These studies have considered incompressible, viscous flows. Another model considered compressible air flow inside insect trachea and tracheoles in the presence of slip[72].

Microfluidic models of insect respiration have also been developed. In a study that sought to recapitulate some of the efficiency of insects in manipulating fluids at the microscale, key features of insect respiratory kinematics, observed via synchrotron-radiation imaging, were incorporated into a series of insect trachea-mimetic microfluidic device designs[73]. Fifteen device designs were fabricated using standard soft lithography techniques. Eleven of the 15 device designs were single-channel, and the remaining 4 device designs were multichannel. All the single-channel devices behaved as microfluidic pumps, converting a single, symmetric actuation pressure input into a unidirectional flow. Furthermore, one of the single-channel device designs exhibited frequency-dependent pumping, wherein the flow reversed direction above a critical actuation frequency. When this direction-reversing single-channel design was incorporated into networked, multi-channel device designs, individual channels in the network could be switched on or off without the use of valves. This finding suggests the possibility of designing valveless microfluidic network architectures that can deliver precise, actuation frequency-dependent control over the flow in individual channels using a single actuation input to the network.

Another series of studies analysed insect-inspired pumping in microfluidic devices induced by either pressure-driven periodic contractions[74,75] or periodic electro-osmotic actuation mechanisms[76,77]. The studies have examined the impact of electro-osmotic phenomena and wall contraction characteristics on various flow quantities, including pressure distribution, velocity distribution, volumetric flow rate pumping, wall shear stress, and streamline patterns, in a high-aspect-ratio micro-tube. One key finding was that thinner electrical double-layer thicknesses led to increased volumetric flow rate and wall shear stresses[77]. In addition, positive Helmholtz–Smoluchowski velocity, U_{HS}, resulted in higher wall shear stress near the membrane, whereas negative U_{HS} led to reversed axial velocity. An overview of peristaltic and membrane-based microfluidic pumping mechanisms is provided in ref. 137. Although electro-osmotic actuation mechanisms require liquids as the flow medium, these studies draw inspiration from insect respiratory actuation patterns, which have the potential to effectively drive liquid flows.

Outlook

Understanding the physical mechanisms behind the diverse respiratory patterns of insects is a complex, interdisciplinary challenge. The intricate interplay between structure and function in insect respiratory systems requires further exploration, and more sophisticated modelling efforts are required to translate this knowledge into usable technology.

Since the advent of X-ray synchrotron imaging in the 2000s, experimental biologists have observed how tracheal collapse and rhythmic contractions facilitate efficient respiration. The impact of key factors, such as tracheal geometry, hydrodynamic slip, spatial and temporal asymmetries, and passive control strategies, has been uncovered. These insights are crucial for understanding how insects maintain high metabolic rates and adapt to varying environmental conditions. However, comprehensive experimental data within the tracheal system are still needed to refine our understanding of these mechanisms. The inability to obtain quantitative measurements or visualizations of rarefied air flows in opaque tracheae remains a key challenge. Without such measurements, models of ventilatory airflow and advective transport in insect respiratory systems cannot be validated.

Although the fundamental principles of insect respiration are robust across species, a more comprehensive understanding of the diversity of insect respiratory strategies and their drivers has not yet

Glossary

Haemolymph
Blood analogue in the open circulatory system of invertebrates, composed of plasma and haemocytes, transporting nutrients, hormones, metabolites and antimicrobial peptides (AMPs) to inhibit bacterial growth.

Knudsen number
In the context of insect respiration, the Knudsen number is defined as the ratio of the molecular mean free path of air to the characteristic length scale of the tracheal system, the tracheal diameter.

Oxidative stress
A condition wherein reactive oxygen molecules exceed the ability of the body to neutralize them, causing potential damage to cellular components such as proteins, lipids and DNA.

Plastron
In aquatic biology, a plastron is an air-retaining structure used for underwater respiration.

Ramifying
A process which refers to branching into successively smaller branches.

Taenidial
Pertaining to taenidia, which are spiral circumferential thickenings on the inner wall of insect tracheae and tracheoles, preventing airway collapse.

Unsteady Reynolds number
Numbers that are often defined for a fluid system with cyclic motions; they encode frequency information and represent the ratio of unsteady inertial forces to viscous forces in the system.

Womersley numbers
Dimensionless numbers for cyclic internal fluid flows that represent the relative importance of transient inertial forces compared to viscous forces in the system; an unsteady Reynolds number.

been formed. The basic insect respiratory plan consists of thousands of ramifying, sometimes collapsible, tracheal tubes that extend all the way to the tissues. This plan may result in distributed, partially automated (via geometric and material properties), and highly variable and tunable (via spiracular control and frequency and intensity of abdominal and locomotor or flight periodic motions) control of nutrient distribution and waste removal. However, variations in tracheal system organization and operation owing to selective pressures are substantial, and further principles await discovery.

To enhance our understanding of these mechanisms, detailed statistical information about tracheal morphometry and collapse kinematics can be obtained from the ever-growing number of live insect-imaging data sets. This information can then be integrated into insect respiration models to understand the influence of asymmetry, inhomogeneity and stochasticity on transport efficiency. The imaging data sets can be used to characterize morphological attributes such as tracheal curvature, cross-section shape, distributions of tracheal diameters, branching angles and morphometric changes in taenidia with tracheal dimensions for various insect species. The data sets can also be used to characterize the timescales and directionality associated with the kinematics of tracheal collapse during RTC. Machine learning methods can help segment these images, easing the computational burden[138,139].

Investigating hydrodynamic slip effects is also essential. As discussed in this Review, respiratory airflow dynamics in the majority of insect tracheae are impacted by their existence in the slip flow regime with appreciable Knudsen numbers, leading to low viscous damping of respiratory gas molecular collisions with each other and with the tracheal wall. By examining local and network-wide hydrodynamic slip phenomena through animal experiments and molecular-level or continuum-slip modelling, researchers can develop a broader understanding of how gas rarefaction effects impact insect respiration and can develop biophysical scaling laws that govern transport in insect respiratory networks.

Another crucial area for improving our understanding of insect respiratory mechanics is determining the correct characteristic timescales in the RTC respiration regime and during flight. Although RTC respiration is typically characterized by the inverse frequency of tracheal collapse, X-ray synchrotron imaging shows that tracheal wall collapse occurs on a much faster timescale. For example, tracheal collapse in *Pterostichus stygicus* happens in hundreds of milliseconds, whereas periodic abdominal compression patterns occur at a mean frequency of 15.6 min^{-1} and last about 2.5 s (ref. 135). An unsteady Reynolds number based on a tracheal wall collapse timescale of 100 ms would be more than two orders of magnitude higher than the equivalent unsteady Reynolds number based on an RTC cycle frequency of 15.6 min^{-1}. This discrepancy suggests that the RTC timescale may need re-evaluation. Doing so would impact current mathematical and computational models, which often assume linearized Stokes flow and quasistatic collapse[65,67–69,71].

During flight, respiration is probably driven by wing flapping motions, which occur at frequencies that can be orders of magnitude higher than typical abdominal ventilatory frequencies in insects. In locusts, for example, wing oscillations are repeated during flight at a frequency of about 20 Hz (ref. 140), whereas ventilatory abdominal contractions during RTC respiration have been recorded at frequencies around 1.5 Hz at atmospheric concentrations of oxygen[141]. Like the case of the disparate tracheal wall collapse and RTC cycle timescales, an unsteady Reynolds number based on a locust wing oscillation frequency of 20 Hz would be more than an order of magnitude higher than the equivalent unsteady Reynolds number based on an abdominal ventilatory frequency of 1.5 Hz. Experimental data support the idea that the wing flapping frequency may be the correct respiratory timescale to consider during flight as well. For example, for the locust *Schistocerca*, the ventilation rate of the tracheal system is 6.25-fold higher when thoracic pumping is added to abdominal pumping during flight[142]. In summary, existing models based on Stokes flow regime assumption may not be accurate and may overlook important, fast-timescale tracheal dynamics.

Advancing our understanding of insect respiration not only contributes to the field of biology but also has broader implications for bioengineering, biomimetics, ecological conservation and other applied sciences because insects can provide a rich source of bioinspiration for tasks involving the manipulation of fluids at the microscale using largely passive control strategies.

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